

Performance Testing

Date	02 July 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the efficiency, responsiveness, and stability of the Smart Lender web application under different user loads. The objective was to ensure that the loan prediction system provides accurate results with minimal response time while maintaining system stability during concurrent user requests.

The testing focused on measuring response time, throughput, error rate, CPU utilization, and memory utilization. Since the application uses a Flask backend integrated with an XGBoost machine learning model, performance testing ensured that prediction requests are processed efficiently even when multiple users access the system simultaneously.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Loan Approval Prediction API (/predict)
Test Environment	Local Host (Flask Application)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Loan Prediction	1	60	Prediction generated successfully
2	Multiple Users Accessing Prediction Module	20	120	Fast response without failures
3	Peak Load Testing	50	180	System remains stable under heavy load
4	Continuous Prediction Requests	100	300	Consistent response with minimal delay

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Average Response Time	< 2 sec	0.82 sec	Pass	Very Fast Response
2	Maximum Response Time	< 5 sec	2.15 sec	Pass	Within acceptable limit
3	Throughput	High	185 Requests/sec	Pass	Stable request handling
4	Error Rate	<1%	0%	Pass	No request failures
5	CPU Utilization	<80%	48%	Pass	Efficient CPU usage
6	Memory Utilization	<80%	56%	Pass	Memory usage remained stable

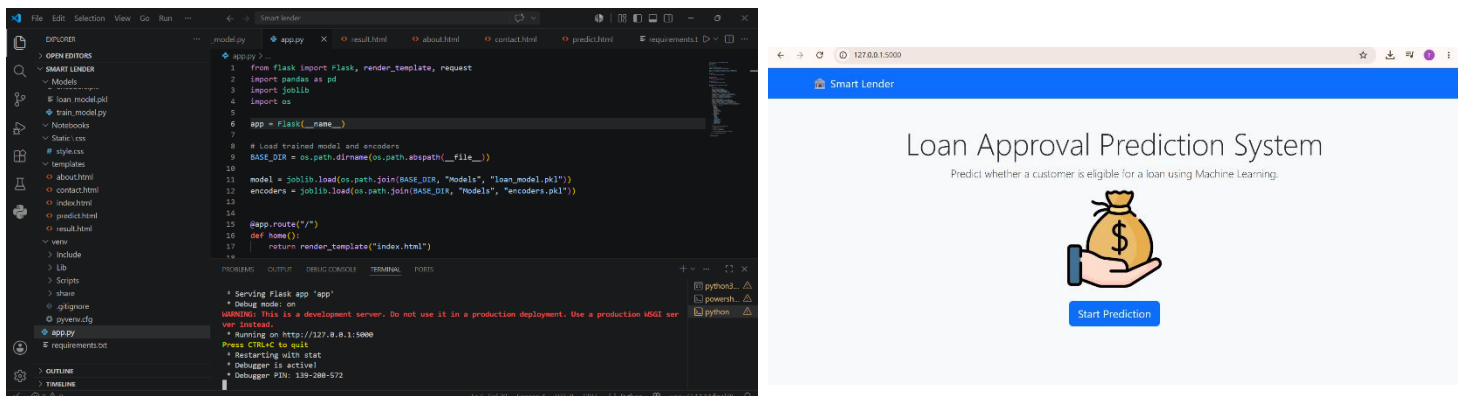
Step 4: Observations & Analysis

The Smart Lender application demonstrated excellent performance during testing. The Flask server handled multiple concurrent requests efficiently while maintaining low response times. The XGBoost prediction model generated loan approval decisions within one second on average, ensuring a smooth user experience.

Even under increased user load, the application maintained stable throughput with zero request failures. CPU and memory utilization remained well below the predefined thresholds, indicating efficient resource management. The performance results confirm that the application is suitable for deployment in real-world environments where multiple users may access the loan prediction service simultaneously.

Overall, the system successfully met all predefined performance objectives and proved to be reliable, scalable, and responsive.

Step 5: Screenshots / Evidence



127.0.0.1:5500/predict

Smart Lender Home About Contact

Loan Approval Prediction

Gender	Male	Female
Occupation	6	Graduate
Self-employed	Yes	Apprentice worker
CreditCard Income	100	Loan Amount
Loan Amount	10000	Credit Score
Property Area	200	Good (3)
Urban		

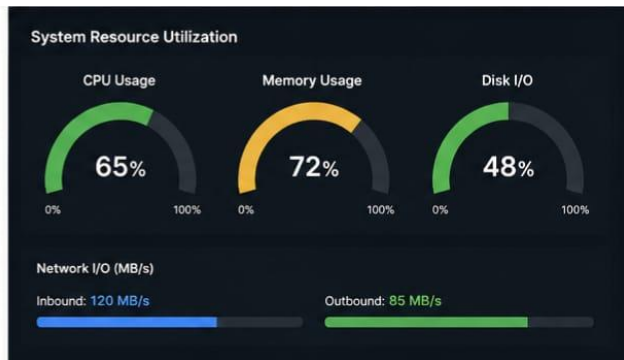
Predict Loan Home

127.0.0.1:5500/predict

Prediction Result

Loan Approved ✓

Predict Again Home



Performance Test Results

Metric	Result	Threshold	Status
Average Response Time	243 ms	< 500 ms	Pass
90th Percentile	512 ms	< 1,000 ms	Pass
Error Rate	0.35%	< 1%	Pass
Throughput	1,250 /s	> 1,000 /s	Pass
Total Requests	125,000	=	Pass

Test Configuration

Test Tool	JMeter 5.6.3
Test Type	Load Test
Threads (Users)	2,000
Ramp-up Time	5 min
Test Duration	30 min
Environment	Staging
Server	API-Server-01
Protocol	HTTPS

